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# MILITARY SPECIFICATION

# INTRAVENOUS INJECTION SET, VENTED, WITH METERING CHAMBER, WITHOUT NEEDLE, DISPOSABLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a vented, terile, disposable intravenous injection set (infusion apparatus) with metering chamber suitable for use in the administration of parenteral solutions from standard glass containers without the use of an airway cannula.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated-Plywood. |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Corner. |
| PPP-B-636 | Box, Fiberboard. |

Military MIL-P-116 Preservation, Methods of

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Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Federal

| | |
|---|---|
| Fed-Std-601 | Rubber: Sampling and Testing. |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-147 | Palletized Unit Loads |
| MIL-STD-794 | Part and Equipment, Procedures for Packaging and Packing of |

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting office.)

2.2 Other publications. The following document form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

AMERICAN IRON STEEL INSTITUTE
Steel Products Manual, Stainless and Heat-Resistant Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 E. 42nd St., NY, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

| | |
|---|---|
| ANSI/HIMA MD 70.1 | American National Standard for Medical Materiel, Luer Taper Fittings, Performance |

(Application for copies should be addressed to the Publications Sales, Dept. STD, American Institute of Physics, 335 E. 45th St., NY, NY 10017.)

U. S. PHARMACOPEIAL CONVENTION, INC.
Pharmacopeia of the United States (U.S.P.).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

NATIONAL INSTITUTE OF HEALTH

Public Health Services Regulations - Part 73*
Pertaining to Federal Control of the Sale, Barter, or Exchange of Biological Products.



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*(Application for copies should be addressed to the laboratory of Biologics Control, National Institutes of Health, Betheda, Maryland 20014.)

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

Federal Food, Drug, and Cosmetic Act and General Regulations for its Enforcement.

(Application for copies should be addressed to the Food and Drug Administration, U.S. Department of Health, Education and Welfare, Washington, DC 20204.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe (nontoxic) when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) of lead when tested as specified in 4.4.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (U.S.P.) for Class IV plastics, when tested as specified in 4.4.11.

3.1.1.1 Molded rubber parts. The molded rubber part shall be fabricated from natural rubber.

3.1.1.2 Plastic tubing. Plastic tubing shall be nontoxic, transparent, colorless polyvinylchloride. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be free from all liquid substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.1.3 Piercing device. The piercing device shall be fabricated from a suitable rigid plastic.

3.1.1.4 Needle adapter. The needle adapter shall be fabricated from a suitable rigid plastic.

3.1.1.5 End protecters. End protecters shall be sheaths or molded caps.

3.1.1.6 Y Injection Site. The Y injection site (medicinal entry assembly) shall be assembled from a part molded of a suitable plastic, a rubber diaphragm stopper, and a stopper securing band or cap.

3.1.1.6.1 Rubber diaphragm. The rubber diaphragm on the medicinal entry assembly leg shall be molded from natural, non-oxidizing, low-sulfur, wax free rubber of 45 $\pm$ 5$^{o}$Shore A Durometer.



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3.1.1.7 Metering chamber. The metering chamber shall be fabricated of 0.012 + .002, - .001 polyvinyl chloride film. The film shall be free from all liquid substances which oxidize, exude or sweat-out to the surface of the film.

3.1.1.8 Roller type control clamp. The roller type control clamp shall be fabricated from a suitable rigid plastic.

3.1.2 Plastic slide clamp. The plastic slide clamp shall be fabricated from rigid plastic. The clamp shall be compatible with the plastic tubing used to fabricate the upper tubing.

3.1.3 Other material. All other materials hereinafter specifed shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Style and design. Style and design shall be in accordance with Figure 1 and as hereinafter specified.

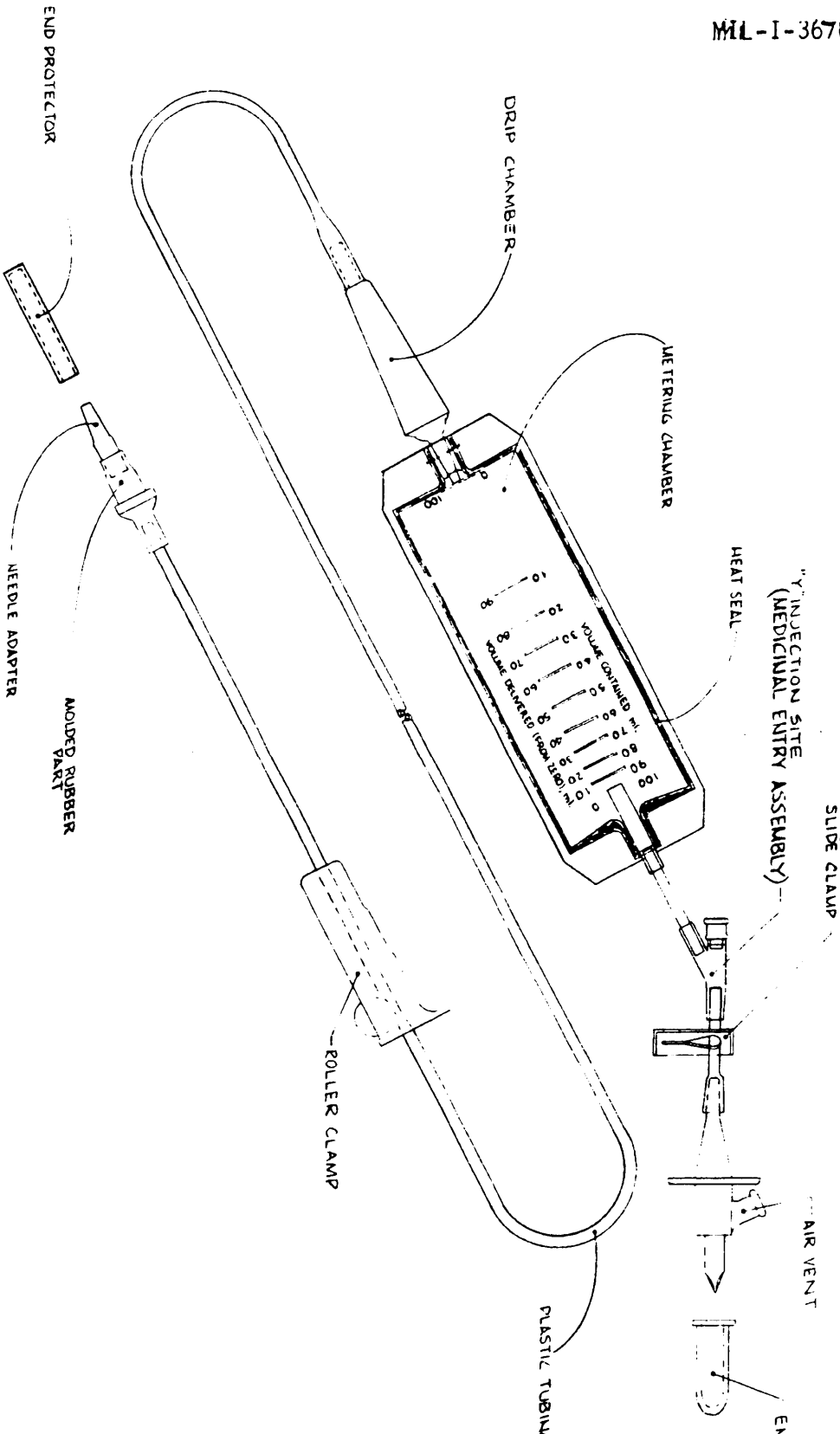
3.2.1 Description. The set (see figure 1) shall be a sterile, non-pyrogenic, nontoxic, disposable intravenous injection set (infusion apparauts) for use in accurate administration of parenteral solutions from standard solution containers (see 3.2.1.2). The set shall consist of a piercing device, a shut off clamp, an in-line graduated metering chamber, a drip chamber, Y injection site, a length of plastic tubing, a molded rubber part, one roller flow control clamp, an air vent with air inletting bacterial filter, a needle adapter, and end protectors. The overall length shall be 73 inches + 3 inches. The piercing device and air vent shall be fabricated from rigid plastic. The assembled set shall be so shaped as to provide a convenient hand grip for the insertion of the piercing device into the stopper or port of the solution container. The assembly of the set shall be structurally sound and shall not leak air or fluid during use. Open ends of the set shall be covered with readily removable plastic covers, or end protectors, which shall maintain thesterility of the fluid path after sterilization. Shall be compatible with the 6515-01-025-8839 infusion pump.

3.2.1.1 Drip chamber. The drip chamber housing shall have an overall length of about 2-7/8 inches, measured from the top of the open drip barrel to the bottom of the tubing conector end. The inside diameter at the bottom stem (where plastic tubing connects) shall be 1/16 + 1/64 inch. The volume capacity of the drip chamber measured up to the level of the stainless steel drip orifice shall be between 3 to 7 ml.

3.2.1.2 Piercing Device (Spike). The piercing device (spike) and air inlet shall be an integrally molded unit. The piercing device shall be a dual-channeled, conical spike, one channel of which shall serve as an air inlet. The piercing device shall be 7/8 to 1 inch in length, with a diameter of 0.258 to 0.267 inch at the base. The piercing device tip shall be sharpened to such an edge as to be suitable for puncture of, and insertion into the stopper or port of a standard solution container (hereinafter specified) without undue pressure (see 3.2.1.2.1). Edges of fluid and air passage openings shall be round or dull to prevent stopper coring. The piercing device shall be ruggedly



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INTRAVENOUS INJECTION SET, WITH METERING CHAMBER, WITHOUT NEEDLE, DISPOSABLE

FIGURE 1



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designed to prevent coring or shearing, and shall be structurally sound. It shall be firmly retained in a leakproof manner when inserted in the stopper or port of a standard solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft under normal storage conditions. The standard solution container stopper referenced above shall be in accordance with requirements contained in the current Defense Medical Purchase Description for NSN 6505-00-975-0604 Dextrose, Injection, USP. The passages in the piercing device shall be properly designed to prevent air from passing directly from the air passage to the fluid passage under maximum flow conditions. The air inlet passage shall vent the solution container (without the need of an airway cannula) during administration of solution. This air inlet opening shall be capped with a removable and replaceable porous air filter which shall prevent the ingress of bacteria (see 3.2.1.2.2) and which shall serve as an air filter during storage and use. The filter cap shall form a snug fit with, and shall be retained on, the opening until manually removed. The fluid passage in the spike shall extend into the drip chamber.

3.2.1.2.1 Durability of piercing device. When tested as specified in 4.4.3 the device shall not loosen, or permit leakage, or become damaged. The piercing device shall be capable of being inserted into the standard solution containers, as described in 4.4.3, without difficulty, the tip (or point) of the device not breaking or bending to the degree of causing difficulty in insertion into standard solution containers.

3.2.1.2.2 Air inletting bacterial filter. The air inletting bacterial filter shall be of such design and material as to comply with the following performance requirements when tested as specified in 4.4.4:

(a) Bacterial retention. The filter shall be such that the rate of contamination of the test media shall be no greater than 0.1% (1-1000) (see 4.4.4.1).

(b) Air passage. The filter shall pass air at the minimum rate of 150 ml per minute (see 4.4.4.2).

(c) Resistance to water. The filter shall support a 30-inch static head of water for 45 seconds with no wetting of the filter element (see 4.4.4.3).



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3.2.1.3 Plastic input tubing. The transparent plastic tubing which connects the spike to the Y injection site shall be about 2 inches long with a nominal inside diameter of 0.135 and a nominal outside diameter of 0.183 inch. The plastic tubing shall be connected to the spike and the single leg of the Y injection site. The attachment shall be tight and leakproof when tested in accordance with 4.4.9. A captive plastic slide clamp designed to close off solution flow shall be positioned on this tubing.

3.2.1.4 Plastic slide clamp. The plastic slide clamp shall be rectangular with a longitudinal opening so tapered and slotted as to provide an open and closed position. The slot (opening) shall be of such dimensions that when the tubing is at the wider end of the taper (open position) the clamp is capable of being rotated 360° around the tube without binding or scoring the tube, and shall allow unrestricted solution flow but also prevent the clamp from sliding below the length of tubing upon which it is assembled. When the tubing is forced into the narrow end of the slot (closed position), fluid flow shall be completely shut off. Exterior corners and edges of the clamp shall be slightly beveled to facilitate the application of pressure required to open or close the clamp. Edges and sides of the slot shall be beveled and smooth enough to prevent the clamp from cutting the tube when in either the open or closed position. The action of opening and closing the clamp shall not cut the tubing when tested in accordance with 4.4.13.

3.2.1.5 Y Injection Site. The appearance of the assembly shall be that of an inverted "Y" so designed that the injection site leg has an internal luer taper in accordance with ANSI/HIMA MD 70.1 and in a straight line with the solution input leg, in order to assure easy and rapid injection of medications. The injection leg shall enter the "Y" at a nominal angle of 30°. The minimum inside fluid passage diameter of the "Y" shall be 0.087 inch.

3.2.1.5.1 Rubber diaphragm. The rubber diaphragm or plug when assembled to the Y injection site shall assure secure retention of the diaphragm upon or plug upon withdrawal of the needle and syringe. After puncture, the diaphragm face shall reseal without a falling drop within 10 seconds when tested as specified in 4.4.12.

3.2.1.5.2 Diaphragm security band. The diaphragm or plug shall be further secured to the entry leg of the Y injection site by a securing band or other retention device forming an airtight, leakproof connection between the diaphragm and the entry leg which shall not separate or leak when tested as specified in 4.4.9.

3.3.1.6 Metering chamber. The metering chamber shall be a bag fabricated by electronically sealing two transparent sheets of 0.012 + .002, - .001 inch polyvinyl chloride film with nominal inside dimensions of 2-1/8 by 5 inches and a volume capacity of 100 ± 5 ml. The chamber shall be connected to the Y injection site at the upper end and directly to the drip chamber at the lower end. The chamber seals shall be a minimum of 1/16 inch wide and of sufficient strength to resist separation or leakage when tested as specified in 4.4.10. The end seals of the chamber shall be bonded to connector bushings. The plastic film shall be free from liquid substances which oxidize, exude or sweat-out to the surface. The chamber shall be imprinted with a double



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volumetric scale graduated in 10 ml increments (0 through 100) indicating both the volume contained and the volume delivered. Graduations shall be accurate within ± 5 cc. Volume contained shall be at the left hand side of the scale as one faces the chamber and the volume delivered at the right hand side. This scale shall be legibly imprinted with a non-toxic ink compounded to bond to the vinyl surface and assure against loss of legibility, due to handling or ordinary contact with standard solutions.

3.2.1.7 Drip chamber. Connected to the output end of the metering chamber is the drip chamber fitted with a calibrated drip orifice. The body of this drip chamber shall be the same as specified in 3.2.1.1. The top connector assembly of the pediatric drip chamber consists of an inlet orifice and stainless steel drip orifice calibrated to deliver 60 ± 9 drops per ml when tested as specified in 4.4.2. The drip chamber shall be directly connected to the bottom of the metering chamber.

3.2.1.8 Plastic tubing. The plastic tubing shall be fabricated of material as specified in 3.1.1.2 and shall have a nominal inside diameter of 0.105 inch and a nominal outside diameter of 0.143 inches. It shall be firmly and permanently secured to the drip chamber and to a molded rubber part at the other end, in such a manner as to form an airtight, leakproof connection which shall not separate when tested as specified in 4.4.9. At the point where the molded rubber part is connected to the plastic tubing, the plastic tubing shall be so designed (or fitted with a thin-walled, rigid plastic reinforcing tube) to prevent the molded part from collapsing the plastic tubing and materially affecting the flow. If a reinforcing tube is used it shall be not less than 1/2 inch long and not less than 0.075 inch in inside diameter.

3.2.1.9 Molded rubber part. The molded natural rubber part shall be a minimum of 1-1/2 inches and a maximum of 2-1/4 inches in length, and shall be constructed with an enlarged center section approximately 5/8 inch in length, and a minimum wall thickness of 0.060 inch. The center section shall have a circumferential shoulder on one and to provide for easy insertion of an 18 gage needle when adding supplemental medication. The molded rubber part shall be capable of resealing subsequent to puncture with a nonsiliconized 18 gage needle cannula. When tested for resealing as specified in 4.4.12 there shall be no falling drop within 10 seconds.

3.2.1.10 Roller type flow control clamp. The control clamp shall be fabricated of plastic and be a roller type clamp from which the roller cannot be accidentally removed. The roller shall withstand a 3.65 kilogram load without being separated from the roller clamp body when tested in accordance with 4.4.5.1. The clamp shall provide a constant setting and quick shut-off from the operating position, shall be movable, and shall be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.5, the clamp shall be capable of accomplishing complete quick shutoff (i.e no falling drop in the drip chamber or from needle adapter (3.2.1.11), shall regulate the flow of the solution from



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complete shutoff to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, the tubing shall not be cut, pierced, or damaged so as to cause leakage or restriction of flow of solution when tested in accordance with 4.4.5.

3.2.1.11 Needle adapter. Needle adapter shall be fabricated of suitable rigid plastic and so constructed as to be attached at one end to the molded rubber part in a leakproof manner. The other end of the adapter shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI/NIMA Standard MD 70.1 in a tight leakproof manner (see 3.7). The adapter, or the assembly of the adapter to the molded part, shall be such as to prevent accidental reversal of the assembly of the adapter to the rubber component at time of assembly. The needle adapter shall not separate from the molded rubber part when tested as specified in 4.4.9.

3.2.1.12 End protectors. End protectors shall be readily removable, transparent or semi-transparent plastic sheaths, or molded caps constructed in such a manner that when placed over each open end of the intravenous injection set it shall, after sterilization, maintain sterility, freedom from pyrogens, and integrity of the entire fluid path of the infusion device until manually removed.

3.3 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, any official of the Federal Food and Drug Administration, or any duly authorized Government representative. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.4 Sterilization, safety, and freedom from pyrogens. All equipment shall be safe (non-toxic). Fluid paths shall be sterile and pyrogen-free. All the above shall be in accordance with current revision of the minimum requirements of the National Institutes of Health for citrated whole blood. The fluid path shall be pyrogen-free when tested as specified in 4.4.6 and sterile when tested as specified in 4.4.7. The sets shall be safe when tested for safety as specified in 4.4.8.

3.5 Certificates. Certificates of sterility, freedom from pyrogens, safety, and bacterial retention as specified below, shall be submitted. For destination inspection, certificates shall be submitted by the contractor directly to Technical Operations Division, Directorate of Medical Materiel, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, PA 19101, Attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection certificates shall be furnished to the procuring activity via the government inspector.



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3.5.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers, shall state that samples from each sterility lot or control have been tested for sterility, as specified in 3.4 and were found to be sterile. The certificate shall reference the method used in determining sterility.

3.5.2 Certificate of freedom from pyrogens. A certificate of freedom from pyrogens, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.5.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

3.6 Compatibility. The materials of which the sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the set which will cause any deleterious effect when the set is used for the purpose intended.

3.7 Leakage and separation of set. Sets shall be tested for leakage and separation of set as specified in 4.4.9. Sets which show leakage or separation at any part of the set shall be rejected.

3.8 Solution flow rate. When tested as specified in 4.4.10, the intravenous injection set, with the filter in its normal dry condition, shall permit the passage of 100 ml of fluid in not more than 5-1/2 minutes, and in not more than 8-1/2 minutes with the filter wet.

3.9 Instructions. A detailed instruction sheet for proper use of the sets, single and in tandom, shall be provided in each unit package.

3.10 Workmanship. Workmanship shall be first class throughout. The intravenous injection sets shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contrct or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastic, metal and rubber, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

Table I. Sampling for Examination.

| Type of defect | Inspection level | AQL (percent defective) |
|---|---|---|
| Major | S-4 | 1.0 |
| Minor | S-4 | 2.5 |

4.2.2 For tests.

4.2.2.1 Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II.

4.2.2.2 For pyrogen, and safety tests. Sampling procedures for pyrogen, and safety tests shall be conducted as indicated in the U.S.P., for Transfusion and Infusion Assemblies.

4.2.2.3 For lead test. Samples shall be taken from each lot of rubber or plastic as indicated in 4.4.

4.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II.



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Table II. Sampling for Other Tests

| Characteristic | Req. | Test Procedure | Insp. level | AQL (percent defective) |
|---|---|---|---|---|
| Class IV Plastic | 3.1 | USP | S-2 | * |
| Lead | 3.1.1 | 4.4.11 | S-1 | * |
| Piercing device durability | 3.2.1.2.1 | 4.4.3 | S-3 | * |
| Bacterial filter | 3.2.1.2.2 | 4.4.4 | 4.4.1 & S-2 | * |
| Plastic slide clamp | 3.2.1.4 | 4.4.13 | S-3 | * |
| Reseal | 3.2.1.5.1 and 3.2.1.9 | 4.4.12 | S-3 | * |
| Drip chamber (Pediatric) | 3.2.1.7 | 4.4.2 | S-3 | * |
| Clamping device | 3.2.1.10 | 4.4.5 and 4.4.5.1 | S-3 | * |
| Needle adapter | 3.2.1.11 | 4.4.9 | S-3 | * |
| Leakage and separation of set | 3.7 | 4.4.9 | S-3 | * |
| Solution flow rate | 3.8 | 4.4.10 | S-3 | * |

* Acceptance number shall be zero for all sample sizes.

4.2.4 Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.2.5 For bacterial retention test. This test shall be performed at least once for each contract. The samples selected shall represent the exact composition, treatment, and construction to be used throughout the entire contract. Any change in the filter shall require additional testing. the number of samples specified in 4.4.4.1 shall be randomly selected from production lots of filters used. Failure of any sample to comply shall be cause for rejection of the filters represented.

4.2.6 For other tests. Sample size for each test shall be as indicated for inspection levels of MIL-STD-105, except that the acceptance number shall be zero for all samples sizes.



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4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Inspection, however, shall not be restricted to the classified possible defects listed in Table III.

Table III. Classification of Defects

| Categories | Defects |
|---|---|
| Major | |
| 101 | Set incomplete. |
| 102 | End protector(s) missing. |
| 103 | End protectors not properly designed to permit flow of sterilizing gases, if used. |
| 104 | End protectors not properly designed to maintain sterility. |
| 105 | Rubber or plastic components tacky. |
| 106 | Style and design not as specified. |
| 107 | Presence of foreign matter. |
| Minor | |
| 201 | Packaging, packing and marking not as specified. |
| 202 | Instructions missing, incomplete, illegible. |
| 203 | Contractor's name or registered trademark, missing, incomplete, or illegible. |

4.4 Tests.

4.4.1 Heavy metals extractable test. A 2 gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm., and eluted for 72 hours at 50°C., 24 hours at 70°C, or 60 minutes at 121°C, in one menstruum of 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml.) of the menstruum. The menstrua are collected, agitated after elution, and then submitted to spectrograhic atomic absorption, spectrophotometric, or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg. from 200 g of test material dispersed in 1,000 ml. of elution menstruum.

4.4.1.1 Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml. of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil for 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml. volumetric flask. Rinse the residue on the filter with deionized water and dilute to 50 ml. Prepare standard solutions of 1, 2 and 5 ppm for each of the metals, namely barium,



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cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadmium, or lead}$$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm, Ba, Cd, or Pb)}}{\text{Absorbance}}$ *

W = Sample weight in grams
* The three standards are averaged to give ppm/unit A.

4.4.2 Drop test. The intravenous injection set shall be tested at room temperature, using water at a rate of 100 + 5 drops per minute. The number of drops required to deliver 2 ml. of water shall be counted, and the drops per ml. calculated from this value.

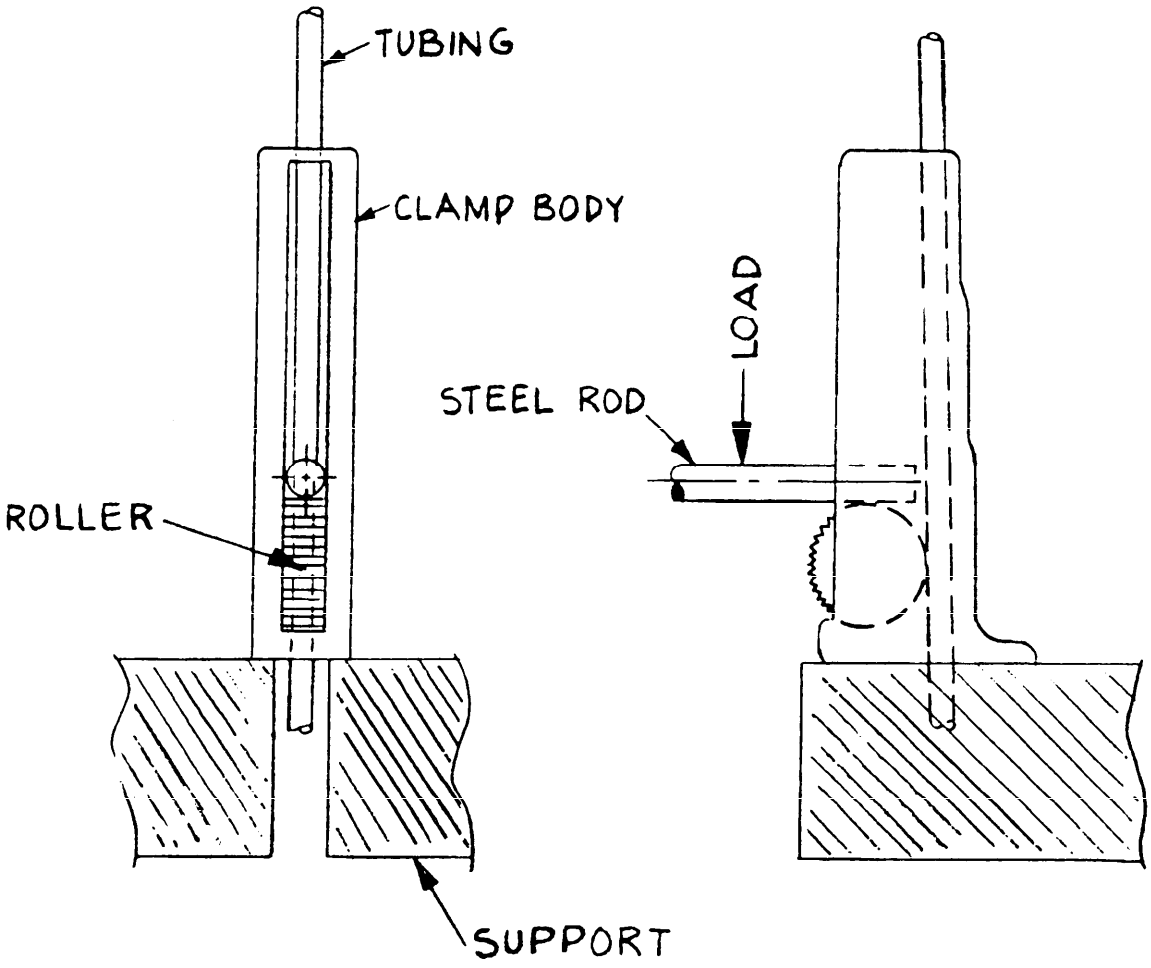
4.4.3 Piercing device durability test. Samples of finished sets, after sterilization, shall be used for this test. The piercing device (on assembled set) shall be inserted into, and removed from, a standard solution container (specified in 3.2.1.2) in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloride solution (in accordance with requirements of the U.S.P. monograph for "Dextrose and Sodium Chloride Injections") for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container as specified in 3.2.1.2. Clamp drip chamber in suitable fixture, apply a five pound load perpendicular to longitudinal axis at 1/8" from sharpened tip, load to be applied for a period of one minute, while still in the fixture apply a five pound load perpendicular to longitudinal axis at a point 1/16" from end of tubing connector for a period of one minute. There shall be no fracture as a result of this test.

4.4.4 Air inletting bacterial filter.

4.4.4.1 Bacterial retention. This following test shall be performed on not less than 50 sample filters randomly selected to represent specific materials used in production batches of filters, and four additional test bottles shall be used for control test for each test period: Detach drip chamber assembly from each sample sterile intravenous injection set by sealing and cutting the plastic tube within 2 inches of drip chamber so that the tubing interior remains sterile. Regular parenteral 1000 ml. (1 liter) bottles having a standard closure shall be filled with 50 ml of fluid thioglycollate medium, U.S.P., evacuated to at least 25 inches of mercury (Hg), sealed and



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METHOD OF APPLYING LOAD TO ROLLER

FIGURE 2



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sterilized. Introduce a sufficient concentration of A. aerogenes into a No. 44 De Vilbiss pocket nebulizer, or similar apparatus. Connect nebulizer to air filter on a drip chamber assembly by means of gum rubber tubing with necessary reducers and connectors. Adjust the concentration of organisms in the nebulizer to obtain a minimum of 300 organisms in the air volume replacing the vacuum in the liter test bottle closure in the normal manner. When the vacuum in the liter test bottle has been displaced, disconnect the testing apparatus from the filter and shake the bottle thoroughly to mix the aerosol content and to wet all surfaces. Retain the drip chamber assembly in the bottle for venting and incubate at $31^{\circ} \pm 1^{\circ}C$ for seven days. As a control to assure that the aerosol is challenging the filter correctly, remove the air filter from the drip chamber assembly of two additional bottles at the start and two additional bottles at the end of each and every test period and connect the nebulizer. When the vacuum has been displaced, disconnect the nebulizer and replace the filter. After mixing the bottles contents, determine the bacterial count of the bottle. The rate of contamination shall be no greater than specified in 3.2.1.2.2.

4.4.4.2 Air passage. The air-inletting bacterial filter shall pass air as specified in 3.2.1.2.2, when tested assembled in the filter cap and under a pressure equal to a column of 16 inches of water. The flow shall be measured with a Brooks Rotameter, or equivalent apparauts.

4.4.4.3 Resistance to water. The air-inletting bacterial filter shall be tested for resistance to water in its filter cap by placing the filter cap on a fitting connected to a 30-inch column of water. The filter shall be examined for wetting at the end of 45 seconds exposure.

4.4.5 Clamping device performance test. The assembled set shall be connected in the normal manner, to a standard solution container as specified in 3.2.1.2 except, as an alternate, the container may be filled with saline solution or water. The container, with set connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit the entire length of the tubing of the set to hang vertically. The ability of the clamp to accomplish complete shut off, to regulate flow, and with a constant head of water maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the molded rubber part. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the needle adapter. The control clamp shall be operated from its fully open position to complete shut off and this operation repeated 10 times while the clamp is in the same position on the tubing and shall meet the requirement specified in 3.2.1.10.

4.4.5.1 Security of roller in clamp body. The roller clamp shall be removed from the set together with a piece of tubing approximately one inch longer than the clamp body. The tubing shall be present in the roller clamp in its normal position with a 1/2 inch length extending from each end of the roller clamp. The clamp (see Figure 2) shall be held vertically, with the



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large end down, over a supporting gap so that the roller if forced out of the body is free to fall. A polished steel tongue is applied to the top of the roller, so that the roller is able to rotate in the course of the test, and the vertical downward force required to separate the roller from the clamp body is measured. As an alternate, any position of the clamp is acceptable providing the clamp body is not distorted and the load measurement is accurate.

4.4.6 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens in accordance with the U.S.P.

4.4.7 Sterility test. The complete intravenous injection set shall be tested for sterility in accordance with the method specified in the U.S.P. using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean -casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30°to 35°C for one half of the specimens and soybean-casein digest at 20°to 25°C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

| Medium | Paper Spore Strip | Inoc.Prod. or Inoc.Simulated Prod. |
|---|---|---|
| Soybean-casein digest | S-3* | S-3* |
| Fluid thioglycollate or Alternative Thioglycollate | | S-3* |

*Use a minimum of 20 specimens. Acceptance number is zero.

4.4.7.1 Inoculated product or simulated product For each type of sterilization, the organism used shall be as listed below:

| Type of Sterilization | Organism |
|---|---|
| Steam | B. stearothermophilus |
| Dry heat | B. stearothermophilus |
| | B. subtilis |
| Ethylene oxide | B. subtilis |
| Gamma radiation | B. pumilus |

The number of viable spores shall be sufficient to provide assurance that the product has been sterilzed.

4.4.7.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586A, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as



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indicated in 4.4.7. Sterility test and 4.4.7.1 Inoculated product or simulated product. the method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.7.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the U.S.P. and paragraphs 4.4.7 and 4.4.7.1.

4.4.7.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

4.4.7.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with the U.S.P. The inspection level shall be as stated in paragraph 4.4.7 above.. Acceptance number shall be zero.

4.4.8 Safety (toxicity) test. The finished sets shall be tested for safety in accordance with U.S.P., for Transfusion and Infusion Assemblies.

4.4.9 Leakage and separation of set. This test shall be applied to those samples subjected to the piercing device durability test (see 4.4.3). The set shall be subjected to an internal air pressure of 4 pounds per square inch (psig) while submerged in a waterbath. No rising bubbles shall be noted in the 20 seconds following application of the pressure.

4.4.10 Solution flow rate. The finished intravenous injection set shall be connected to a standard container as specified in 3.2.1.2 in the normal manner, the entire assembly inverted and suspended so that there is a 30 inch column of fluid in the set and container and no restriction of flow imposed by the clamp. The time necessary for passage of 100 ml. of fluid through the entire set shall be determined. This test procedure shall be repeated after purposely submerging the air filter in distilled water for a period of 10 minutes.

4.4.11 Biological test. The plastics shall be tested in accordance with the U.S.P. for Class IV plastics. Sampling will be as shown in Table II.

4.4.12 Reseal test. The rubber diaphragm and molded rubber part shall reseal without a falling drop within 10 seconds after puncture with a nonlubricated 18 gage needle when subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. for 30 seconds.

4.4.13 Slide clamp test. The slide clamp shall be operated for 10 complete cycles from stop to full flow with the clamp located approximately in center of tubing length while subjected to an internal pressure (hydrostatic pressure) of 4 p.s.i.g. At the completion of the test the tubing shall be examined for cuts.



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4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with Groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the Quality Assurance Provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Unit package. Each intravenous injection set shall be packaged in a box of appropriate size and design. The tubing shall be coiled in such a manner to prevent kinking or taking a permanent set. The box design shall include suitable interior fittings to prevent damage to contents. Closure of the box shall be adequate to prevent spilling of contents under normal handling. The box shall be further packaged in a sealed bag conforming to MIL-P-116, method IC-2.

5.1.1.2 Intermediate package. Forty-eight intravenous injection sets shall be packed in a box conforming to PPP-B-636, class domestic. Closure shall be effected with tape or adhesive as specified in the box specification. Metal stitches shall not be used.

5.1.2 Level C. The level C preservation for injection sets shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Three intermediate packages (144 unit pgs) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V30. Closure and strapping shall be as specified in the applicable box specification.

5.2.2 Level B. Three intermediate packages (144 units PGs) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.



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5.2.4.1 Levels A and B. Intravenous injection sets packed as specified in 5.2.1 and 5.2.2, shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Intravenous injection sets packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B, and C.

5.3.1.1 Unit package. Each package shall be marked in accordance with commercial practice, requirements of the Federal Food, Drug and Cosmetic Act, and MIL-STD-129. In addition, each unit shall contain instructions as specified in 3.9, and the legend: "Fluid path and end connectors are sterile and pyrogen free. Set is ready for use." In addition, markings shall include the lot or control number. Date of manufacture (month and year) shall be shown in lieu of date packed.

5.3.1.2 Each intermediate pack, exterior container and unitized load shall be marked as specified in MIL-STD-129. Markings shall include the lot or control number. Date of manufacture shall be shown in lieu of date packed.

5.4 General.

5.4.1 Exterior container. (5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock number items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of parenteral solutions in controlled volume from 10 ml to 100 ml at 60 drops per cc from standard solution bottles.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

| | |
|---|---|
| NSN 6515-00-088-8868 | INTRAVENOUS INJECTION SET, Vented, With Metering Chamber, Without Needle, Disposable, 72 inches |



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6.4 Supersession data. This specification includes the requirements of Purchase Description No. 2, dated 29 September 1969 for NSN 6515-00-088-8868, INTRAVENOUS INJECTION SET, With Metering Chamber, Without Needle, Disposable.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing Activity:
Defense Supply Agency - DM

Project No. 6515-3277

Review Activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination for changes to this document, draft circulation should be based on the information in the current DODISS.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-I-36708A | INTRAVENOUS INJECTION SET, VENTED, WITH METERING CHAMBER, WITHOUT NEEDLE, DISPOSABLE |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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